

The point here is that in such scenarios, technologies tend to stack over each other to form—not a delicious sandwich with everything in it—but a leaning tower of impending doom, a technological time bomb sitting on a fault line. Such has been the legacy of the Windows family of products: bandage old technologies with just enough new code to not break the old while clumsily introducing the new. Windows Vista aims to address this ill of Windows with swooping changes made to several internals of the OS, the least of which, gaining the most attention, is the User Interface (UI) itself. But before we do that, we need to take a quick look at the various changes that Vista has introduced at the back-end of the operating system.

2.2.1 A View To The Future

Back in the olden days, people gasped in amazement over the marvel that was the VGA card: it brought colour and resolutions unheard of. Spreadsheets were suddenly more fun, and games, more real and dangerous. Before we get into further details, let's quickly understand what most operating systems are built around: APIs, or Application Programming Interfaces, are a bunch of instructions that tell a program how to interact with another program. An operating system is generally a bunch of such interconnecting interfaces—each telling the other how to talk to files, or draw windows, or print your documents, and so on. One such API is used to draw everything that you see on screen in a Windows environment. Called the GDI (Graphics Device Interface), the API is a remnant from 1993—back from Windows NT 3.1. The API saw several layers added to it, which brought in more features and greater complexities to programmers.

2.2.2 The Limitations Of A Bitmap Interface

The GDI is a bitmap-based drawing tool for the operating system. When it draws elements on screen, it creates actual images much like one would in Paint or Photoshop. The problem with creating interface elements using pixels is that if those elements are required to say be resized or animated, the image loses its quality